

-:Neuromorphic Computing:-

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STUDENT CODE:- BWU/BTA/22/232

COURSE NAME:-Technical Seminar With Modern Tool Application

Course Code:-PROJ-CSM681

DEPARTMENT NAME:- CSE(AI&ML)

SEC:- D , SEM:- 6



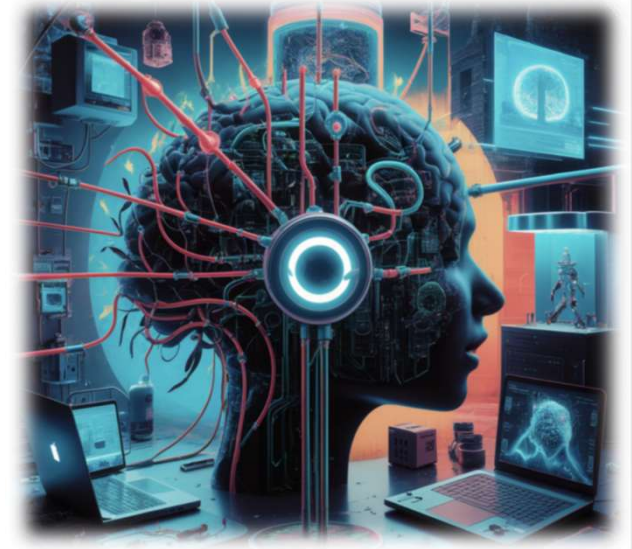
Content:-

1. Introduction to neuromorphic computing.
2. Historical Background.
3. Key principles & Key features.
4. Application.
5. Advantages.
6. Future Directions.
7. Conclusion.

Introduction to Neuromorphic Computing

Neuromorphic computing is a brain-inspired approach to computing that uses artificial neurons and synapses.

It aims to create hardware that processes information similarly to the human brain.

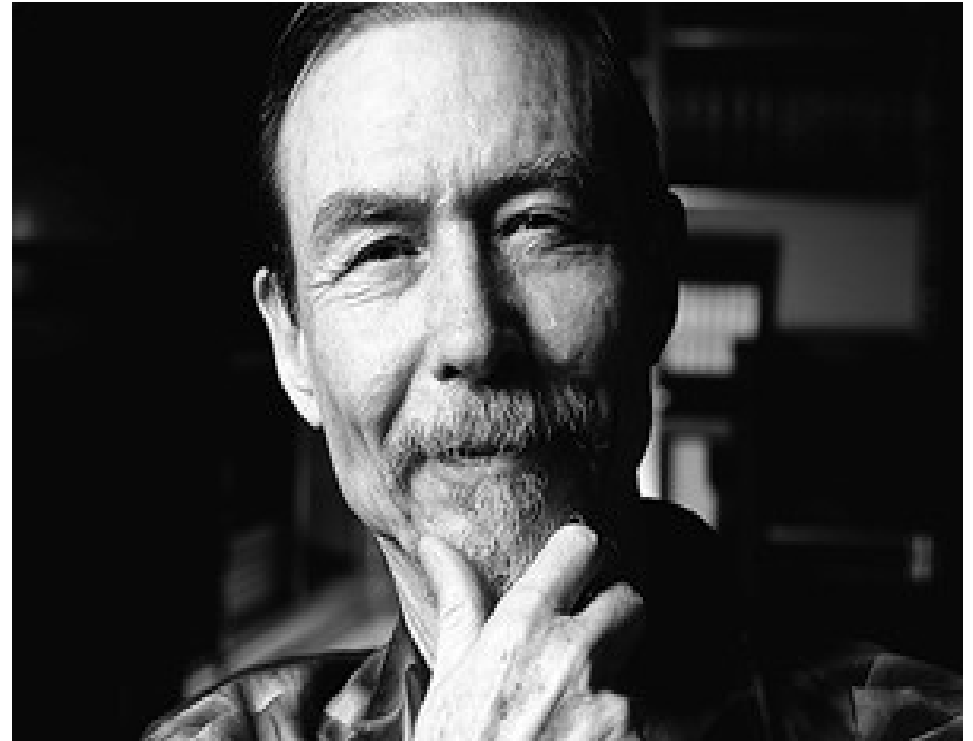


Historical Background

The concept of neuromorphic engineering was first introduced by Carver Mead in the late 1980s.

Early projects focused on developing analog circuits that emulate neural activity.

Advances in digital technology have expanded capabilities and applications over recent decades.



Key Principles and Features

Includes spiking neurons, synapses, and event-driven communication mechanisms.

These systems are inherently parallel, enabling efficient processing of complex data.



Application of neuromorphic computing

Neuromorphic computing enhances real-time sensor processing in autonomous robots. It enables energy-efficient implementations of machine learning tasks.

Used for:-

- Driverless cars
- Drones
- Robots
- Smart home devices
- Natural language, and image processing

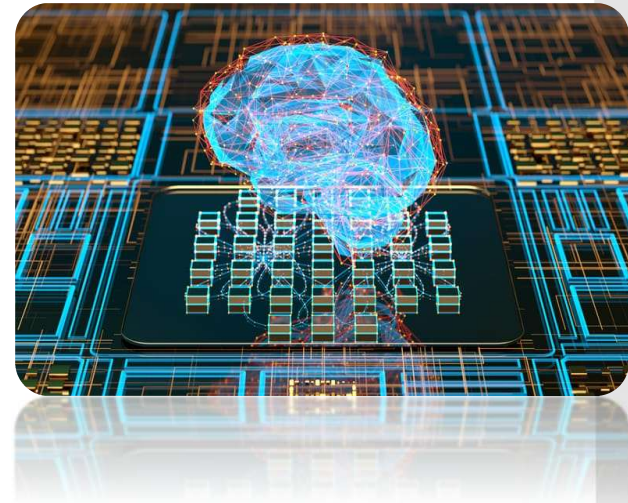


Advantages:-

Neuromorphic systems consume significantly less power compared to conventional CPUs and GPUs.

Energy-efficient, scalable, adaptable, and highly parallel.

On-chip learning and adaptation and Real-time decision making

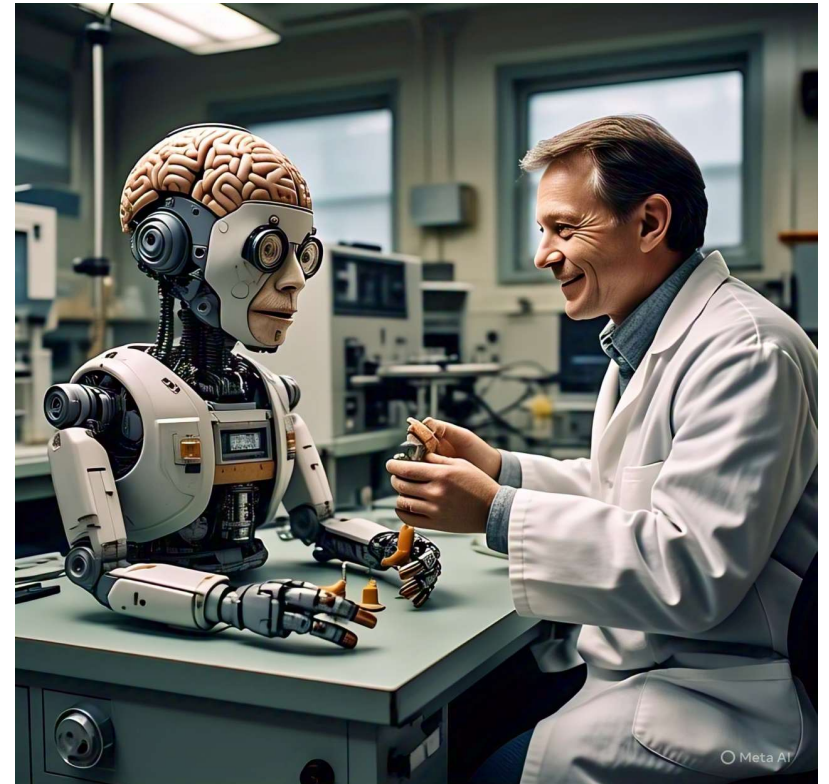


Future Directions

Researchers are exploring hybrid systems combining neuromorphic hardware with traditional computing.

Advances in materials like memristors promise more efficient synaptic elements.

The field aims to achieve more autonomous, adaptive, and brain-like AI systems.



Conclusion:-

Neuromorphic computing is the most favorable paradigm to realistically replicate the computing processes in the human brain which features the highly parallel and event-driven scheme of computation. Also, the brain uses energy only when and where it is needed for processing the information.



THANK YOU